E-CaTCH

Event-centric cross-modal attention
with temporal consistency and class-imbalance handling
for misinformation detection

Ahmad Mousavi
Department of Mathematics and Statistics, American University

Joint work with Yeganeh Abdollahinejad, Roberto Corizzo,
Nathalie Japkowicz, and Zois Boukouvalas

Department of Mathematics, Georgetown University
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The story is larger than one post

- A caption can read as ordinary until the picture is attached.
- A real photograph can still be paired with a false claim.
- Many posts carry one story, and the wording shifts over days.
- A score on a single post never sees that shift.

Misinformation here is multimodal, and it is a sequence.

Three gaps in post-level detectors

1. Joining a text vector to an image vector does not decide which modality to trust when they disagree.
2. Models that score posts one by one miss how a narrative drifts inside an event.
3. Real posts dominate. High accuracy can hide a miss on the rare class.

The repair has three mathematical pieces: a gate, a short sequence of window vectors, and a weighted loss.

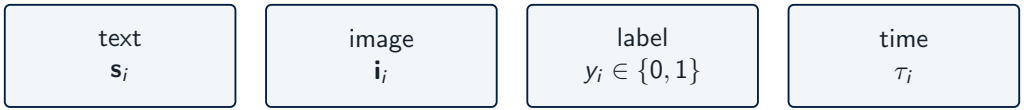
E-CaTCH reads an event, then a trend

Group related posts into a *pseudo-event*. Fuse text and image with a coordinate-wise gate. Read the event as a short sequence of overlapping time windows.

1. Keep both cross-modal directions, then mix them with a gate.
2. Track the change from one window to the next.
3. Train with class weights, a temporal penalty, and ℓ_2 regularization.

Plan for the hour: the vocabulary, the model, the three datasets, then the evidence.

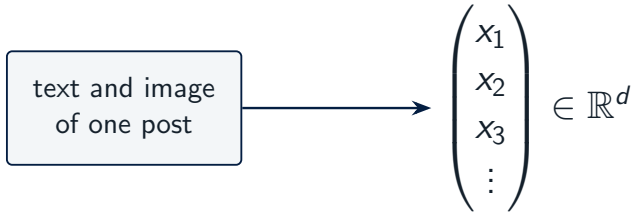
A post is four things



$$(\mathbf{s}_i,\ \mathbf{i}_i,\ y_i,\ \tau_i)$$

- y_i is a post label: real or fake. The event label is not given.
- τ_i is the timestamp. Order is part of the data, not a feature we invent later.

An embedding turns a post into a vector



- The coordinates are features, not words and not pixels.
- Posts the encoder treats as similar land near each other.
- Text and image each get a vector. A later linear map puts them in one shared $\mathbb{R}^d$.

Two frozen encoders, then a linear map

BERT reads the text

A pretrained text encoder. We do not train it from scratch, and it is not ChatGPT.

We keep one vector, the `[CLS]` embedding, in $\mathbb{R}^{768}$.

ResNet-152 reads the image

A pretrained image network. We keep the vector before its classifier, in $\mathbb{R}^{2048}$.

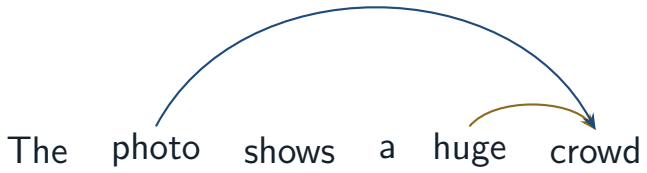
A missing image becomes the zero vector, so every post has the same shape.

A learned linear map sends both vectors into one dimension d:

$$\mathcal{T}_i = W_{\text{text}}\,\mathrm{BERT}_{\text{CLS}}(\mathbf{s}_i) + b_{\text{text}}, \qquad \mathcal{I}_i = W_{\text{img}}\,z_i + b_{\text{img}}.$$

$W_{\text{text}} \in \mathbb{R}^{d\times 768}$ and $W_{\text{img}} \in \mathbb{R}^{d\times 2048}$. The encoders stay fixed; these two maps are trained.

Self-attention: each word looks at the others



- "Huge" has to be licensed by "crowd." "Photo" is pointing at the same noun.
- Each word builds a new vector as a weighted sum of the others.
- The weights are the attention. They depend on the words themselves.

One formula: scaled dot-product attention

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

- Q, K, and V are learned linear maps of the same sequence.
- Row i, column j of $QK^\top$ is a score: how much token i uses token j.
- Dividing by $\sqrt{d_k}$ keeps those scores from saturating the softmax.
- **Multi-head** means several of these maps in parallel. Their outputs are concatenated and mixed by one more matrix.

Cross-attention changes where the query comes from

Text looks at the image

The query comes from the text.
Keys and values come from the image.

"The streets are empty."
The picture is a packed square.

Does the picture support the words?

The image looks at the text

The query comes from the image.
Keys and values come from the text.

The same packed square.
The same sentence about empty streets.

Do the words describe this picture?

Both questions get an uncomfortable answer. We keep both answers.

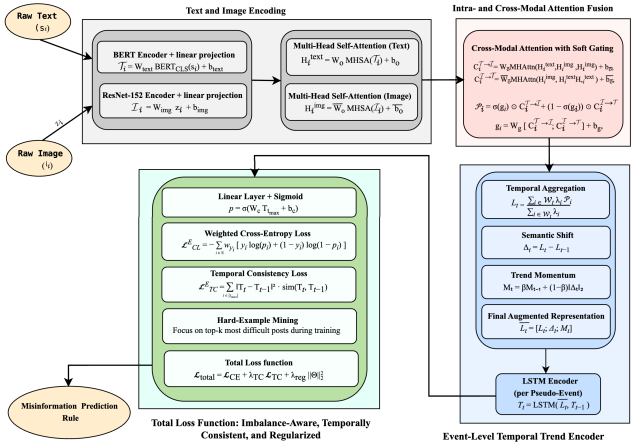
The gate chooses which direction to trust

Attention has already built two stories: what the text sees in the image, and what the image sees in the text.

- A weight near 1 keeps the text-looks-at-the-image story.
- A weight near 0 keeps the image-looks-at-the-text story.
- The weight is a *vector*. Different coordinates may trust different modalities.
- The weight is chosen per post, from the two stories themselves.

Alignment finds the conflict. The gate decides the mix.

Four blocks, then a decision



Encoders, gated fusion, the temporal trend, and the loss. Figure from Mousavi et al., IEEE TCSS 2026.

Events are clusters we build, not labels we are given

- Standardize sentence embeddings to mean zero and variance one.
- Agglomerative clustering, cosine distance, average linkage, *inside each split*.
- Choose the number of clusters on validation, by event-level F_1.
- Images are used after the clusters exist. They do not define the clusters.

Clustering on Fakeddit	Silhouette ↑	Davies–Bouldin ↓	Intra-sim. ↑
Text only (BERT)	0.37	1.05	0.64
Image only (ResNet)	0.26	1.36	0.46
Text and image, concatenated	0.32	1.18	0.57

Text wins every column. Images are often reused, missing, or weakly tied to the caption. The absolute scores are modest because the stream is noisy; the comparison is the evidence.

The gate is a convex combination in each coordinate

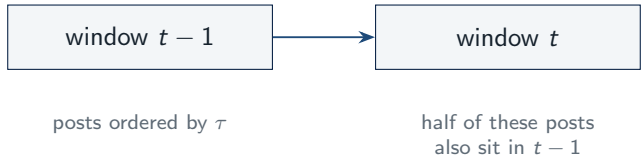
Each modality vector is passed through the self-attention map, with that same vector as query, key, and value. Call the results H_i^{text} and H_i^{img}.

$$C_i^{\mathcal{T}\to\mathcal{I}} = W\,\mathrm{MHAttn}(H_i^{\text{text}}, H_i^{\text{img}}, H_i^{\text{img}}) + b, \quad C_i^{\mathcal{I}\to\mathcal{T}} = \overline{W}\,\mathrm{MHAttn}(H_i^{\text{img}}, H_i^{\text{text}}, H_i^{\text{text}}) + \overline{b}.$$

$$\mathcal{P}_i = \sigma(g_i) \odot C_i^{\mathcal{T}\to\mathcal{I}} + \big(1 - \sigma(g_i)\big) \odot C_i^{\mathcal{I}\to\mathcal{T}}$$

$\sigma(g_i) \in (0,1)^d$, so coordinate k of $\mathcal{P}_i$ is a convex combination of the two directions. The gate is $g_i = W_g \begin{bmatrix} C_i^{\mathcal{T}\to\mathcal{I}} \\ C_i^{\mathcal{I}\to\mathcal{T}} \end{bmatrix} + b_g$, with $W_g \in \mathbb{R}^{d\times 2d}$. With one vector on each side, the interesting mix is this gate.

Inside an event, time is a row of overlapping windows



- Default overlap $\rho = 0.5$: the stride is half a window, so neighbors share half their posts.
- The clock depends on the dataset. Fakeddit windows run about 3–5 days. The other two are set with their own data slides.
- Overlap keeps a gradual shift from falling into the crack between two windows.

Recent posts count more, and we record the jump

Inside window t

$$\lambda_i = \exp\big(-\alpha(\tau_{t,\max} - \tau_i)\big)$$

$$L_t = \frac{\sum_{i \in \mathcal{W}_t} \lambda_i \mathcal{P}_i}{\sum_{i \in \mathcal{W}_t} \lambda_i}$$

The newest post has weight 1. Older posts in that window shrink. $\alpha > 0$.

From window $t-1$ to t

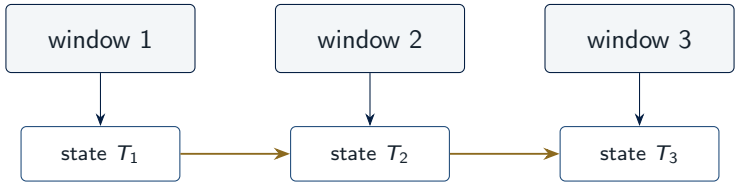
$$\Delta_t = L_t - L_{t-1}$$

$$M_t = \beta M_{t-1} + (1-\beta)\,\|\Delta_t\|_2$$

Δ_t is the change in the window summary. M_t tracks the size of that change. $0 < \beta < 1$.

The LSTM receives $\bar{L}_t = [L_t; \Delta_t; M_t] \in \mathbb{R}^{3d}$, with the scalar M_t repeated across d coordinates.

An LSTM carries a state from window to window



- The event is a short sequence of windows. One new vector updates the state.
- The state and the cell start at 0, and both are reset at every new event.
- We classify from T_{T_E}, the state after the last window.

One probability per event, with a class weight

$$p_E = \sigma(W_c T_{T_E} + b_c)$$

The weight of class $c \in \{0, 1\}$ is computed on events, not on posts:

$$w_c = \frac{\bar{n}}{n_c + \varepsilon}, \qquad \bar{n} = \frac{n_0 + n_1}{2}, \qquad \varepsilon > 0.$$

$$\mathcal{L}_{\mathrm{CE}} = -\sum_{E \in \mathcal{E}} w_{y_E} \big[y_E \log p_E + (1 - y_E) \log(1 - p_E) \big].$$

The smaller class gets the larger weight. Training also emphasizes events whose loss is already large.

The event label comes from the last window

$$y_E = \mathbf{1}\left(\sum_{i \in \mathcal{W}_{T_E}} \lambda_i \, y_i \;\geq\; \frac{1}{2} \sum_{i \in \mathcal{W}_{T_E}} \lambda_i \right)$$

- Only the most recent window votes, and recent posts inside it vote more.
- Early speculation in the same event does not get a vote.
- The displayed rule uses $\geq$. The paper also says a tie defaults to label 0.

Event-level annotation is not in the data. This is the supervision the loss sees.

The loss asks for fit, smoothness, and small weights

$$\mathcal{L}_{\mathrm{TC}} = \sum_{E \in \mathcal{E}} \sum_{t=2}^{T_E} \| T_t - T_{t-1} \|_2^2 \cdot \mathrm{sim}(T_t, T_{t-1})$$

sim is cosine similarity. The penalty is the squared step times the cosine of the angle between neighboring states.

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{TC}} \mathcal{L}_{\mathrm{TC}} + \lambda_{\mathrm{reg}} \|\Theta\|_2^2$$

Θ is the set of trainable parameters. Each term is computed at event level.

Fakeddit: a large Reddit stream, with no event labels

- About 1.2 million Reddit posts, text and image, real or fake.
- Topics are mixed: politics, entertainment, and much else.
- Events are not annotated. Pseudo-events come from the clustering step.
- Inside a cluster, posts are ordered by time and cut into windows of about 3–5 days.
- Overlap $\rho = 0.5$.
- Misinformation is the minority class, as it is on the other two sets.

IND: fact-checked images from Indian elections

Fact-Checked Images Shared During Elections.

- Campaign material from 2013–2021. Each item is a claim plus an image, real or fake.
- Moderately sized, and strongly imbalanced: misinformation is about 20% of the samples.
- Daily windows, overlap $\rho = 0.5$, in step with a political news cycle.
- Explicit event labels are not provided, so the same pseudo-event construction applies.

MediaEval 2020: COVID-19 tweets

The MISINFOGRAPH task from MediaEval 2020.

- Pandemic-related tweets, labeled misleading or accurate.
- Mostly text, with some images.
- Weekly windows, overlap $\rho = 0.5$.
- Health narratives move more slowly than election news, so the window is longer.
- Misinformation is again the minority class.

Three platforms, three clocks, one overlap rule.

One protocol on all three datasets

- Split 80/10/10 into train, validation, and test.
- Clustering and windowing are fit *inside* each split, so posts from one story do not leak across the cut.
- Every number below is a mean $\pm$ standard deviation over 5 seeds.
- Hyperparameters are chosen on validation only. The test set is scored once.
- When two model variants are compared, they share encoders, windows, optimizer, and class weights. Only the piece under study changes.
- Misinformation is the minority class on all three. That is the realistic setting, and the class weight is there because of it.

The same model on three datasets

Dataset	Accuracy (%)	Precision	Recall	F_1	AUC
Fakeddit	95.5 ± 0.30	0.957 ± 0.006	0.953 ± 0.009	0.955 ± 0.008	0.975 ± 0.004
IND	89.8 ± 0.36	0.903 ± 0.009	0.890 ± 0.011	0.896 ± 0.010	0.925 ± 0.005
COVID-19	89.5 ± 0.38	0.899 ± 0.007	0.885 ± 0.010	0.891 ± 0.009	0.938 ± 0.005

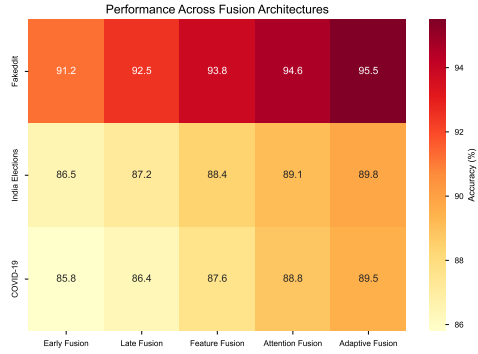
Fakeddit is the largest and most mixed, and it is where the model is strongest. IND and the COVID-19 set are harder, and both are minority-class problems. The paper reports a 1.57 percentage-point accuracy gain on Fakeddit over prior strong baselines.

On Fakeddit, against published multimodal models

Model	Accuracy (%)	Precision	Recall	F_1
EANN	87.5	0.9043	0.8811	0.8926
MVAE	88.8	0.9011	0.9139	0.9074
ELD-FN	88.8	0.9354	0.9029	0.9189
MMBT	91.1	0.9274	0.9251	0.9263
MTTV	91.9	0.9348	0.9303	0.9325
BMR	91.7	0.9434	0.9288	0.9361
GAMED	93.9	0.9355	0.9371	0.9363
E-CaTCH	95.5	0.9570	0.9530	0.9550

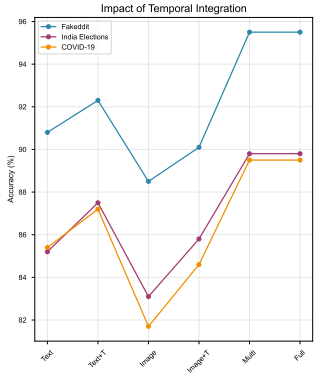
EANN adversarial; MVAE generative; ELD-FN event-level fusion; MMBT early fusion; MTTV temporal transformer; BMR multi-view; GAMED multi-expert. Same table as the paper. Best entry in each column is ours.

Adaptive fusion is best on every dataset



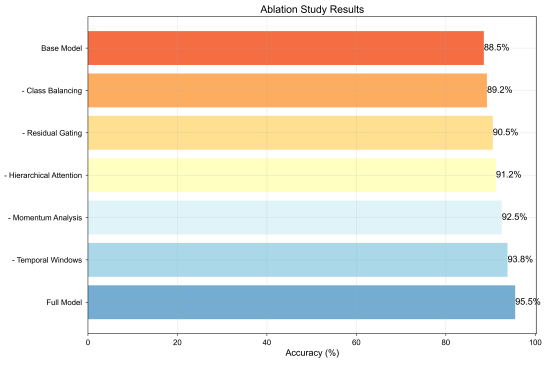
Only the fusion block changes. Early fusion to adaptive fusion, as printed: $91.2 \to 95.5$, $86.5 \to 89.8$, $85.8 \to 89.5$.

Time lifts each modality; both together sit at the top



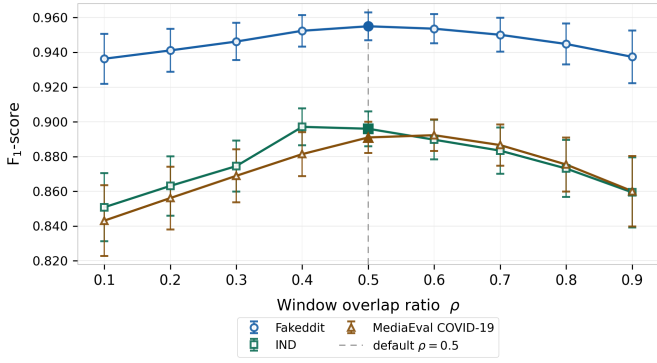
Text, text plus time, image, image plus time, multimodal, full model. On each curve the image-only point is lowest, time lifts text and lifts image, and the multimodal and full models sit together above either modality.

Removing a module lowers accuracy



Each intermediate bar disables one module. The figure prints a base model at 88.5% and the full model at 95.5%. Class balancing removed: 89.2%. Temporal windows removed: 93.8%. Class balancing is the adaptive weight; residual gating is the soft gate.

Overlap near one half is a stable default



F_1 stays stable for ρ from 0.4 to 0.6. Fakeddit peaks at 0.5. IND is slightly higher at 0.4, and COVID-19 at 0.6; both gaps from 0.5 sit inside the run-to-run deviation. The extremes 0.1 and 0.9 are weaker. The main experiments use $\rho = 0.5$.

Train on one platform, test on another

No fine-tuning. Hyperparameters stay at the source dataset.

Train $\rightarrow$ test	Accuracy (%)
Fakeddit $\rightarrow$ IND	88.2 ± 0.40
Fakeddit $\rightarrow$ COVID-19	87.8 ± 0.42
IND $\rightarrow$ Fakeddit	89.5 ± 0.38
COVID-19 $\rightarrow$ Fakeddit	89.1 ± 0.39

A model trained on Reddit still lands near 88% on election images and on COVID-19 tweets.

Where the method is thin

- Clustering uses text only. Memes and posts with almost no caption can be split across events.
- A point-in-time event gives the temporal branch little signal. The model then falls back to a stable event vector.
- Audio, video, external knowledge, social graphs, and a measured inference cost are open.

The gate is a per-post reading of which direction was trusted. Using that reading as a display for fact-checkers is future work, not a result in this paper.

Three things to keep

1. The unit is the event we build from text and time, not the isolated post.
2. Keep both cross-modal directions. Mix them with a convex combination in each coordinate.
3. A short sequence of windows, with decay, shift, and momentum, is enough time structure when the loss respects class imbalance.

On that design: 95.5% accuracy on Fakeddit, 89.8% on IND, and 89.5% on the COVID-19 set.

Questions

I would like to leave the rest of the hour for questions.

- The gate, the event label, and the text-only clustering are the pieces I am happiest to unpack.

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Code: github.com/Yegi03/E-CaTCH